

Hamromaster

COMPLETE NOTES

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**For Hamromaster Online
Classes**

**Whatsapp:
9840842566**



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Analysis of categorical Data.

Simple classification



manifold classification

Notation of attributes

Positive sense

→ Presence → A, B, C, D etc

Negative sense

→ absence of attributes → a, b, c, d or α, β, γ .

Class frequency (AB) / ($\alpha\beta$)

(A) = Bracket ले बाहरकी ओर class frequency हुआ।

both positive ओर → (AB) positive class frequency.

Negative ओर → ($\alpha\beta$), ($\beta\gamma$)

(A α) → Complementary ← (C, r)
↓
वर्ग
class frequency.

order of class

A $\rightarrow$ 1st order

AB $\rightarrow$ 2nd order

ABC $\rightarrow$ 3rd order



$n = 3^n$ ✓

① For two attributes A and B

	A	α	Total
B	(AB)	(αB)	(B) =
β	(A β)	($\alpha\beta$)	(β) =
Total:	(A)	(α)	

$$(A) = (AB) + (A\beta)$$

$$(\alpha) = (\alpha B) + (\alpha\beta)$$

$$B = (AB) + (\alpha B)$$

$$\beta = (A\beta) + (\alpha\beta)$$

$$N = (A) + (\alpha)$$

$$N = (B) + (\beta)$$

(2)

Yule's coefficient of Association (युलेंज)

$$Q = \frac{(AB)(\alpha\beta) + (A\beta)(\alpha B)}{(AB)(\alpha\beta) + (A\beta)(\alpha B)}$$

Value of Q lies between -1 and $+1$.
It is interpreted like coefficient of correlation.

Yule's coefficient of colligation

$$Y = \frac{1 - \sqrt{\frac{(AB)(\alpha\beta)}{(AB)(\alpha\beta)}}}{1 + \sqrt{\frac{(AB)(\alpha\beta)}{(AB)(\alpha\beta)}}}$$

$$Q = \frac{2Y}{1+Y^2}$$



Pearson's coefficient of Contingency

$$C = \sqrt{\frac{\chi^2}{N + \chi^2}}$$

palpali bharat

where, $\chi^2 = \frac{N[(AB)(\alpha\beta) - (\alpha\beta)(A\beta)]^2}{(A)(\alpha)(B)(\beta)}$ (i)

or $\chi^2 = \sum \frac{(O-E)^2}{E}$ (ii)

$\chi^2 = \sum \frac{(O-E)^2}{E}$ (iii)

Ultimate class frequency

→ In the classification of units of a population, according to certain attributes, the classes of the highest order are called the ultimate classes. The frequency of any ultimate class is called ultimate class frequency.

for example:

$$(A) = (AB) + (AP)$$

$$(A) = (ABC) + (ABP) + (APR)$$



Note: Consistency of data

यदि table मा ~~सर्व~~ value
(+) भए consistency यदि
(-) भए not consistency
हो ।

Methods of measures of Association

- (i) frequency method
- (ii) proportion method
- (iii) Yule's coefficient of association
- (iv) Yule's coefficient of colligation
- (v) Pearson's coefficient of contingency.

1. for two attributes A and B, we have
 $(AB) = 35$, $(A) = 55$, $N = 100$ and $(B) = 65$
 calculate remaining class frequencies.

solⁿ

Here, Given,

$$(AB) = 35$$

$$(A) = 55$$

$$N = 100$$

$$(B) = 65$$

Prepare the 2×2 table using relations
 between class frequency and
 class frequencies.

	A	α	Total
B	$(AB) = 35$	$(\alpha B) = 30$	$(B) = 65$
β	$(A\beta) = 20$	$(\alpha\beta) = 15$	$(\beta) = 35$
Total	$(A) = 55$	$(\alpha) = 45$	$N = 100$

the remaining frequencies are -

$$(\alpha) = 45$$

$$(A\beta) = 20$$

$$(\alpha\beta) = 15$$

$$(\alpha B) = 30$$

$$(\beta) = 35$$

Sol ⁿ	A	α	Total
B	$(AB) = 130$	$(\alpha B) = 40$	$(B) = 120$
β	$(A\beta) = -50$	$(\alpha\beta) = 330$	$(\beta) = 280$
Total	$(A) = 80$	$(\alpha) = 320$	$N = 400$

Since, all class frequency are not positive, Hence the given data are not consistent or inconsistent.

- (4) ~~Test~~ Given $N = 820$, $(A) = 250$, $(\alpha B) = 50$ and $(AB) = 35$. Test whether the data is consistent.

Solⁿ
we have,

$$N = 820$$

$$(A) = 250$$

$$(\alpha\beta) = 50$$

$$(AB) = 35$$

We representing the above information into 2×2 Contingency table:

	A	α	Total
B	$(AB) = 35$	$(\alpha\beta) = 50$	85
β	$(A\beta) = 215$	$(\alpha\beta) = 520$	735
Total	250	570	820

(5)

Now, for the given data to be, consistent we must have

$$(AB) \geq 0, (A\bar{B}) \geq 0, (\bar{A}B) \geq 0, (\bar{A}\bar{B}) \geq 0$$

Again from question we can see that

$$(AB) \geq 0, (A\bar{B}) \geq 0 \text{ and } (\bar{A}B) \geq 0$$

~~Now,~~

~~As~~ These is no negative ultimate frequency class, the given data is consistent.

2.6) If there are 144 A's and 384 B's in 1024 observations, how many AB's will be there, A and B being independent.

Solⁿ

Here,

Given,

$$(A) = 144$$

$$(B) = 384$$

$$N = 1024$$

$$\therefore (AB) = \frac{(A) \times (B)}{N}$$

$$= \frac{144 \times 384}{1024}$$

$$= 54 \text{ Ans}$$

} independent case की लागू है यी।

Q.9) In experiment on the immunization of cattle from Tuberculosis the following results were obtained. ^{रोग}

	Cattle (गाइयतु)	
	Affected	Not affected
Inoculated	6	13
Not inoculated नो-इनोकुलेटेड	8	13 (book की ans मिलान की जाती है इससे)

Using Yule's coefficient of association study the relationship between inoculation and exemption from Tuberculosis.

Solⁿ

Here,

Let A, α , B & β are denoted by affected not affected, inoculated and not inoculated respectively.

T.U 2072

	A	α	Total
B	(AB) = 6	(α B) = 13	(B) = 19
β	(A β) = 8	(α β) = 13	(β) = 21
Total	(A) = 14	(α) = 26	N = 40



Yule's coefficient of Association

$$Q = \frac{(AB)(\alpha\beta) - (A\beta)(\alpha B)}{(AB)(\alpha\beta) + (A\beta)(\alpha B)}$$

$$= \frac{6 \times 13 - 8 \times 13}{6 \times 13 + 8 \times 13} = \frac{-26}{182} = -0.1428 \text{ Ans}$$

Q. No. 11

(a) Investigate the association between darkness of eye colour in father and son from the following data.

Fathers with dark eyes and Sons with dark eyes = 50

Father with dark eyes and Sons without dark eyes = 79

Father without dark eyes and Son with dark eyes = 89

Fathers without dark eyes and Sons without dark eyes = 782

(b) What would have been the frequency of fathers with dark eyes and sons with dark eyes for the same total number, had there been complete independence.

So?

Here,

(a) $\Rightarrow$ Let $A \rightarrow$ fathers dark eyes
 $\alpha \rightarrow$ father without dark eyes
 $B \rightarrow$ Son's dark eyes
 $\beta \rightarrow$ Son's without dark eyes

According to this question

$$(AB) = 50 \quad (\alpha\beta) = 782$$

$$(A\beta) = 79$$

$$(\alpha B) = 89$$

$$\therefore g = \frac{(AB)(\alpha\beta) - (A\beta)(\alpha B)}{(AB)(\alpha\beta) + (A\beta)(\alpha B)}$$

$$= \frac{50 \times 782 - 79 \times 89}{50 \times 782 + 79 \times 89}$$

$$= \frac{39100 - 7031}{39100 + 7030}$$

$$= \frac{32069}{46130}$$

$$= 0.695 \quad \underline{\text{Ans}}$$



b)

Now,

/

⑥

	A	α	
B	$(AB) = 50$	$(\alpha B) = 89$	$(B) = 139$
β	$(A\beta) = 79$	$(\beta\alpha) = 782$	$(\beta) = 861$
Total	$(A) = 129$	$(\alpha) = 871$	$N = 1000$

Now,

Complete independence

$$= \frac{(A) \times (B)}{N}$$



$$= \frac{17931}{1000}$$

$$= 17.9$$

$$\approx 18 \text{ Ans}$$

अंश

12 In a sample study of deafness in a locality, the following figures were obtained:

	Sane ^{मानसिक}	Insane ^{अमानसिक}	Total
Deaf	20 ^(A)	40 ^(B)	60
Not deaf	50	25	75
Total	70	65	135

Trace the intensity of association between sanity and deafness. Also calculate the Colligation & verify with the χ^2 relation formula.

Solⁿ

Here,

Let (A), (α) and (B), (β) be the Same, insane and deaf, not deaf respectively. Then,

	Same (A)	insane (α)	Total
Deaf (β)	(AB) = 20	($\alpha\beta$) = 40	(B) = 60

Not deaf (β)	(A β) = 50	($\alpha\beta$) = 25
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Total	(A) = 70	(α) = 65
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(B) = 75

N = 135

Now,

$$Y = \frac{1 - \sqrt{\frac{(A\beta) \times (\alpha\beta)}{(AB) \times (\alpha\beta)}}}{1 + \sqrt{\frac{(A\beta) \times (\alpha\beta)}{(AB) \times (\alpha\beta)}}}$$

$$= \frac{1 - \sqrt{\frac{50 \times 40}{20 \times 25}}}{1 + \sqrt{\frac{50 \times 40}{20 \times 25}}} \Rightarrow Y = -0.6$$

Again,

The relation between Q and Y is given by

$$Q = \frac{2Y}{1+Y^2} = \frac{2 \times (-0.6)}{1+(-0.6)^2} = -0.33$$

$$\therefore Q = -0.33$$

Pearson's coefficient of contingency:

It is connected with chi-square test. The coefficient of contingency is worked out when chi-square test confirms the association between attributes. Coefficient of contingency is given

by formula,

$$c = \sqrt{\frac{\chi^2}{N + \chi^2}}, \text{ where } \chi^2 = \frac{\sum (O - E)^2}{E}$$

Thus, the Pearson's coefficient c lies between 0 & 1 but never attains the value unity. The value of c lies near 1 shows every high degree of association between the two attributes and value near to zero show no association. ✓

A driving school examined the results of 100 candidates who took their test for the first time. It was found that out of 40 men, 28 passed and out of 60 women, 34 passed. Calculate (i) Yule's coefficient of colligation (ii) Pearson's coefficient of Contingency?

Solⁿ

Here,

Let A, α, B & β denoted by men, women and pass and failed respectively.

	(M) A	α (W)	
$B(P)$	$AB = 28$	$(B\alpha) = 34$	$(B) = 62$
$P(F)$	$(A\beta) = 12$	$(\alpha\beta) = 26$	$(\beta) = 38$
	$(A) = 40$	$(\alpha) = 60$	$N = 100$

• Yule's coefficient of ~~colligation~~ ^{Association}

$$Q = \frac{(AB)(\alpha\beta) - (A\beta)(\alpha B)}{(AB)(\alpha\beta) + (A\beta)(\alpha B)}$$

$$= \frac{28 \times 26 - 12 \times 34}{28 \times 26 + 12 \times 34}$$

$$= \frac{728 - 408}{728 + 408} \Rightarrow \frac{320}{1136} = 0.2816$$



• Yule's coefficient of colligation

$$Y = \frac{1 - \sqrt{\frac{(A\beta) \times (\alpha B)}{(AB) \times (\alpha\beta)}}}{1 + \sqrt{\frac{(A\beta) \times (\alpha B)}{(AB) \times (\alpha\beta)}}}$$

$$= \frac{1 - \sqrt{\frac{12 \times 34}{28 \times 26}}}{1 + \sqrt{\frac{12 \times 34}{28 \times 26}}} \Rightarrow 0.144$$

Again,

$$\chi^2 = \frac{N[(AB) \times (\alpha\beta) - (\alpha\beta) \times (A\beta)]^2}{(A) \times (\alpha) \times (B) \times (\beta)}$$

$$= \frac{100[28 \times 26 - 26 \times 12]^2}{40 \times 60 \times 62 \times 38}$$

$$\Rightarrow 3.0605$$

$$\therefore \chi^2 = 3.061$$

pearson's coefficient of contingency

$$c = \sqrt{\frac{\chi^2}{N + \chi^2}} \Rightarrow \sqrt{\frac{3.061}{100 + 3.061}}$$

$$= 0.132$$

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